Kola Govind Mani

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.04
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh (BIEAP))	2020	9.67%
Kesava Reddy School	Class X (Board of Secondary Education, Andhra Pradesh (BSEAP))	2018	9.80%

EXPERIENCE

Indo MIM Ltd: New Product Development (NPD) Engineer

- **Modelled** the part, runner, and gate in **CREO** and **Assembled** the full casting tree to import into MAGMAsoft simulation software.
- **Simulated** metal pouring and cooling in **MAGMAsoft** across temperatures and fill times to spot defects like shrinkage and porosity.
- **Redesigned** gates and runners in CAD to fix metal flow issues and reduce casting defects.

Liquid Propulsion Systems Centre (LPSC - ISRO): Research Intern (Samudrayaan Project)

- **Modelled** a *3D cross-section of a liquid propulsion tank* in **MSC Apex** for **Electron Beam Welding (EBW)** analysis.
- **Designed** a **DOE** by varying beam power and welding speed to study their effect on weld penetration.
- **Analysed** Ti-6Al-4V **EBW** using **Finite Element Analysis** in **Simufact Welding** to study thermal and thermo-mechanical behaviour.
- **Identified 20 kW and 2 mm/s** as the optimum condition for **100 mm DOP** with acceptable stress and distortion.

IIT Tirupati: Research Intern (Hybrid Manufacturing)

- **Troubleshot IRO 3D** machine and slicing issues during the initial setup; worked with a **US client** during off-hours.
- **Designed** copper components in **SolidWorks** and sliced them using **IRO 3D software** for printing.
- **Printed** copper components using the **IRO 3D hybrid additive manufacturing system**, with silica sand as the support material.
- **Studied** sintering and holding time conditions and identified **950°C for 90 minutes** as optimum for **densification and bonding.**

SKILLS

Software: SolidWorks, CATIA, Ultimaker Cura, Abaqus CAE, MSC Apex, Simufact Welding, IRO 3D, ANSYS Workbench, ANSYS Fluent.

FEA: Structural Analysis, Thermo-mechanical Analysis (Welding & Additive Manufacturing simulations), Mesh Generation and Refinement, Heat Source Modelling, Hyper elastic Material Modelling.

Manufacturing: Additive Manufacturing (FDM and Metal AM), Electron Beam Welding (EBW), Reverse Gravity Casting, VPIC/EQ Casting.

Programming: Fortran (User Subroutines), MATLAB, Python (Basic Automation).

RELEVANT COURSES

Design for Metal Additive Manufacturing (DfAM), Advanced Finite Element Methods (AFEM), Computational Fluid

Dynamics (CFD) Tools, Machine Learning and its Applications, Design for Manufacturing and Assembly (DFMA), Industry 4.0

PROJECTS

Finite Element Analysis of Residual Stress in LPBF Components

- **Modelled** different **AM components** in **SolidWorks** and generated G-code in **Ultimaker Cura** with different process parameters.
- **Developed** a **Python script** to extract and interpolate G-code coordinates for the laser tool path.
- **Generated and refined meshes** using **HyperMesh**, then matched **element centroids** with interpolated G-code coordinates using a **Python-based bounding-box approach**.
- **Applied DFLUX** in **Abaqus** to control heat-source movement along the G-code path and compared **residual stresses** with experimental results.

Nano-Scale Tensile Load Analysis of PDMS-Gold Particle System Using FEA

- **Modelled** a nanoscale **PDMS-Gold particle system** in **Abaqus** using **Ogden hyperelastic model** to study stress under tensile loading.
- **Compared** stress and **gold-particle displacement** with experimental results using **combined-part** and **Tie-constraint** models.
- **Found 2% error** with the **combined-part model** versus **50% error** with the **Tie-constraint model** after **mesh refinement**.

CFD Analysis of Thermal Mixing in a Y-Junction with Passive Flow Control

- **Modelled** a 3D **Y-junction** and simulated thermal mixing of hot and cold water streams using **ANSYS Fluent**.
- **Analysed** the effects of **inlet angle, baffle geometry, and baffle location** on mixing efficiency under laminar flow conditions.
- **Achieved 96% mixing efficiency** using **curved-surface baffles** placed at **1D and 3D** downstream of the junction.

Multi-Track Surface Melting of IN718 Using FEA

- **Modelled** a 3D **IN718 substrate** in **Abaqus** and simulated two adjacent laser melt tracks with **30%, 40%, and 50% overlap**.
- **Implemented** a **Goldak heat source** using the **DFLUX** subroutine and used **USDFLD** to track **melt pool and HAZ**.

Design and Fabrication of a Gyroscopic Cup Holder Using Additive Manufacturing

- **Designed** a functional two-axis **gyroscopic cup holder** in **Solid Edge** for FDM printing.
- **Applied DFAM principles** and adjusted the **build orientation, slicing parameters, and supports** to reduce build time.
- **Fabricated** the assembly using **FDM with PLA** on a **Creality Ender 3 V3** across two build plates in **12 h 08 min**.

SCHOLASTIC ACHIEVEMENTS

- **Top 5 Outgoing Student, Mechanical Engineering (2024):** Ranked **5th** in Mechanical Engineering, JNTUACEA.
- **Elite + Silver Medal - NPTEL:** Scored **87%** in *Introduction to Mechanical Micro Machining*, ranking in the **Top 2%**.

POSITION OF RESPONSIBILITY

- **Teaching Assistant (TA),** *IIT Hyderabad*: Assisted in teaching, laboratory sessions, student support, and academic evaluation.
- **PG Buddy** *Sunshine, IIT Hyderabad:* **Supported** fellow postgraduate students through peer interaction and guidance.
- **Teaching Volunteer**, *Akshara Dhan Teaching Initiative:* **Teaching and supporting** Class 10 students in their academic learning.

EXTRACURRICULARS

Interested in Squash, Cricket, Watching Movies, Learning New Things, and Internet Awareness.